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GitHub Link: https://github.com/DanSadeh/fNIRS-Signal-Processing

Theoretical Background and Methodology Neural activity requires oxygen, which is delivered to active brain regions via the vascular system. By shining near-infrared light into the tissue and measuring its attenuation, we can track these hemodynamic changes and estimate the concentrations of oxygenated (HbO) and deoxygenated (HbR) hemoglobin. Because light scatters through biological tissue rather than traveling in a straight line, the standard Beer-Lambert law is insufficient. We therefore use the Modified Beer-Lambert Law (MBLL), which accounts for the physical scattering using a Differential Pathlength Factor (DPF).

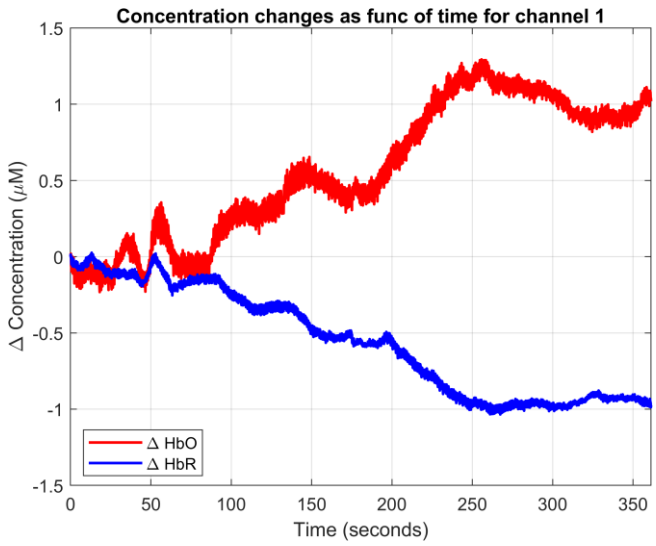
In the first step of our data processing (within the CalcNIRS function), we convert the raw, unitless light intensity—which exhibits exponential decay—into Optical Density (OD) using a base-10 logarithm. This allows us to solve the system linearly: $OD = \log_{10}\left(\frac{I_0}{I_t}\right)$

For the baseline intensity (I_0), we selected the intensity at the first index (time = 0) without averaging across multiple samples. Because task-based hemodynamic responses and cardiac cycles occur as rapid, dynamic changes, the initial state provides a clean, stable reference point for the tissue prior to any induced cognitive load, making a multi-sample average unnecessary.

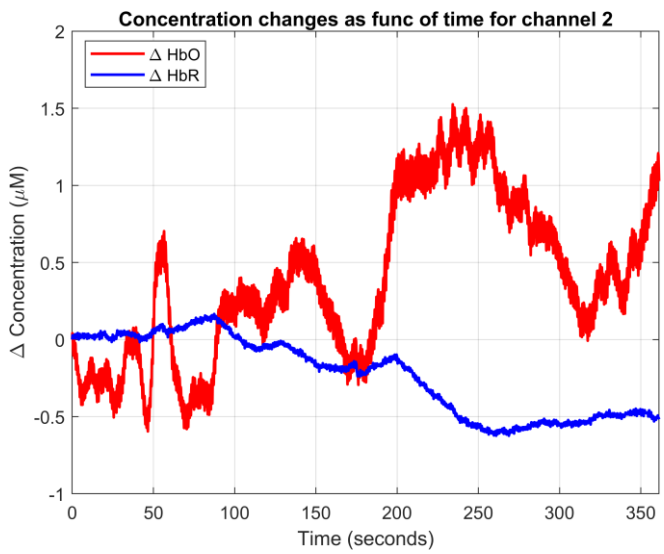
After computing the OD, we calculate the effective pathlength (L_{eff}) by multiplying the physical source-detector separation by the base DPF and the wavelength-dependent relative coefficient, $L_{eff} = SDS \cdot DPF \cdot relDPF$. Finally, we extract the concentration changes (ΔC) in microMolar (μM) by multiplying the distance-normalized OD by the inverted extinction coefficient matrix:

$$\Delta C = \varepsilon^{-1} \cdot \frac{OD}{L_{eff}}$$

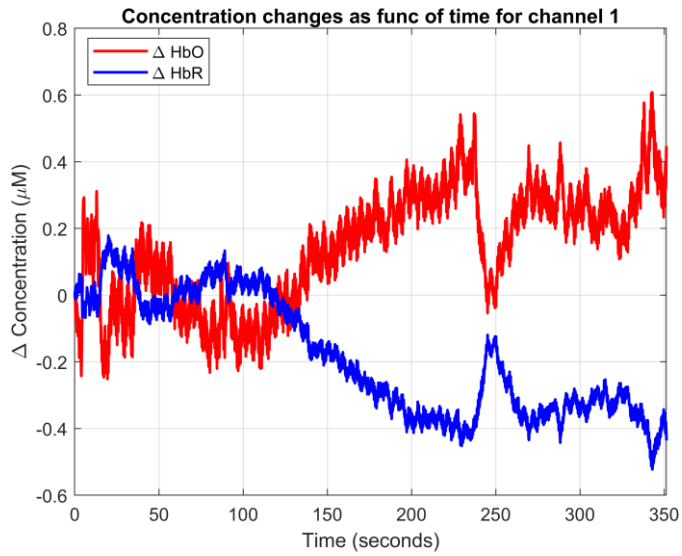
Time Domain Concentration Changes (Question 2) Applying the MBLL equations allowed us to successfully unmix the overlapping absorption spectra. We first analyze the data from recording session 1 (File 1: FN_032_V1). The plot below displays the concentration changes over time for Channel 1. As expected physiologically, the ΔHbO and ΔHbR signals fluctuate inversely within the microMolar range, reflecting the influx of oxygen-rich arterial blood displacing oxygen-depleted blood.



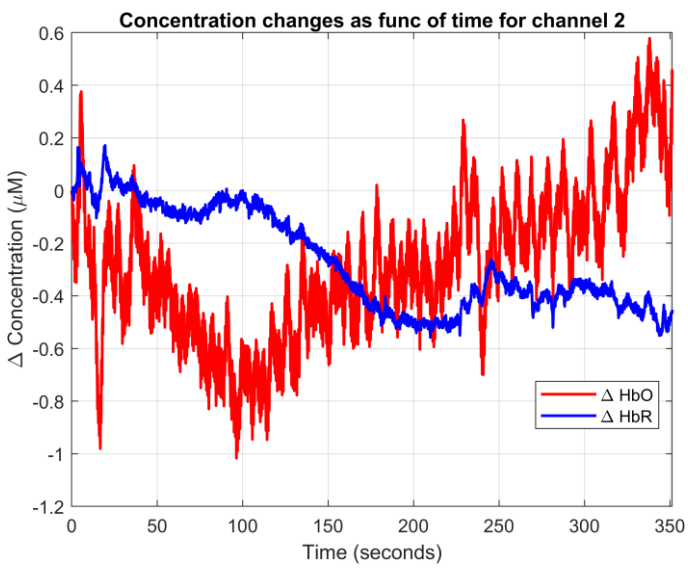
Similarly, looking at Channel 2 from the same recording session (File 1), we observe the same inverse hemodynamic relationship, confirming the stability of the MBLL matrix inversion across different spatial optode locations.



Moving to the second recording session (File 2: FN_031_V2), the plot below shows the time-domain concentration changes for Channel 1. The dynamic range remains consistent with physiological expectations in the microMolar scale.



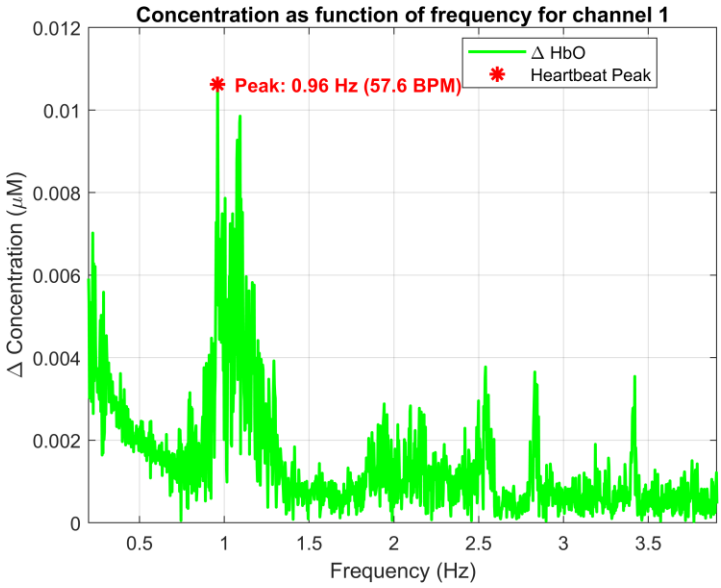
Finally, the plot for Channel 2 in the second recording session (File 2) is presented below, fully demonstrating the algorithm's capability to continuously track localized blood flow changes over time.



Frequency Domain Analysis and Signal-to-Noise Ratio (Question 3) To isolate the physiological cardiac cycle from system noise, we transformed the time-domain ΔHbO signal into the frequency domain using a Fast Fourier Transform (FFT). We analyzed the oxygenated signal specifically because the physical shockwave of the heartbeat is most prominent in fresh arterial blood.

After normalizing the energy to preserve true physical amplitudes, we mathematically removed the 0 Hz DC offset. The plot below is displayed starting from 0.2 Hz to intentionally exclude massive, slow-moving physiological artifacts (such as respiration or Mayer waves) that would otherwise skew the visual scale. As explicitly marked on the graph with a red indicator, the algorithm

successfully isolated a distinct cardiac peak at a resting heart rate of 57.6 BPM. The calculated Signal-to-Noise Ratio (SNR) is 14.07, proving that the physiological heartbeat is vastly stronger than the high-frequency (> 2.5 Hz) hardware noise floor.



File: EX1__314631763.m

Matlab

```
% Dan Sadeh 314631763
clear;
clc;
close all;

%% Calculation segment - Initialization and Constants
% Defining hardware and physiological constants upfront to avoid magic numbers
SOURCE_DETECTOR_SEP = 3;
HEARTBEAT_LOW_HZ = 0.8;
HEARTBEAT_HIGH_HZ = 1.5;
NOISE_FLOOR_HZ = 2.5;
MICROMOLAR_CONV = 1e6;
SECONDS_PER_MINUTE = 60;

% Defining inputs
dataFile1 = '.\FN_032_V1_Postdose1_Nback.mat';
dataFile2 = '.\FN_031_V2_Postdose2_Nback.mat';
tissueType = 'adult_head';
plotChannelIdx = [1, 2];
extinctionCoefficientsFile = '.\ExtinctionCoefficientsData.csv';
DPFperTissueFile = '.\DPFperTissue.txt';
relDPFfile = '.\RelativeDPFCoefficients.csv';

%% Calculation segment - Time Domain Processing (Question 2)
% Applying the MBLL to both recording sessions to extract relative concentrations
[dHbR_1, dHbO_1, fig_1] = CalcNIRS(dataFile1, SOURCE_DETECTOR_SEP, tissueType,
plotChannelIdx, ...
    extinctionCoefficientsFile, DPFperTissueFile, relDPFfile);

[dHbR_2, dHbO_2, fig_2] = CalcNIRS(dataFile2, SOURCE_DETECTOR_SEP, tissueType,
plotChannelIdx, ...
    extinctionCoefficientsFile, DPFperTissueFile, relDPFfile);

% Automatically exporting Question 2 Time Domain plots at 600 DPI
exportgraphics(figure(fig_1(1)), 'File1_Channel1_TimeDomain.png', 'Resolution', 600);
exportgraphics(figure(fig_1(2)), 'File1_Channel2_TimeDomain.png', 'Resolution', 600);
exportgraphics(figure(fig_2(1)), 'File2_Channel1_TimeDomain.png', 'Resolution', 600);
exportgraphics(figure(fig_2(2)), 'File2_Channel2_TimeDomain.png', 'Resolution', 600);

%% Calculation segment - Frequency Domain Transformation (Question 3)
% Extracting the time vector to determine the hardware sampling rate for the FFT
dataStruct1 = load(dataFile1);
timeVec = dataStruct1.t;
samplingFreq = 1 / mean(diff(timeVec));
numSamples = length(timeVec);

% Subtracting the mathematical mean eliminates the 0Hz DC offset
centeredSignalChannel1 = dHbO_1(:, 1) - mean(dHbO_1(:, 1));

% Executing the discrete Fourier transform
```

```
fftResultHbO = fft(centeredSignalChannel1);

% Normalizing by the total sample count ensures the Y-axis reflects physical
amplitude
fftAbsHbO = abs(fftResultHbO / numSamples);

% Slicing the array to discard negative frequencies
fftFinalHbO = fftAbsHbO(1:floor(numSamples/2) + 1);

% Multiplying by 2 conserves the physical energy
fftFinalHbO(2:end-1) = 2 * fftFinalHbO(2:end-1);

% Constructing the physical X-axis
freqAxis = (0:floor(numSamples/2)) * (samplingFreq / numSamples);

%% Calculation segment - SNR & Heart Rate
% We calculate the peak BEFORE plotting so we can explicitly mark it on the graph
heartBeatMask = (freqAxis >= HEARTBEAT_LOW_HZ & freqAxis <= HEARTBEAT_HIGH_HZ);
[heartBeatStrength, maxIdx] = max(fftFinalHbO(heartBeatMask));

% Extracting exact frequency to convert to Beats Per Minute (BPM)
maskedFreqs = freqAxis(heartBeatMask);
heartBeatFreq = maskedFreqs(maxIdx);
bpm = heartBeatFreq * SECONDS_PER_MINUTE;

% Defining the baseline hardware noise floor
noiseMask = (freqAxis >= NOISE_FLOOR_HZ);
noiseAvg = mean(fftFinalHbO(noiseMask));

% Final ratio calculation
SNR = (heartBeatStrength / noiseAvg);

%% Plot segment - FFT Spectrum Visualization with Peak Marker
fig_fft = figure;
% Magnifying signal to microMolar to view biological changes clearly
plot(freqAxis, fftFinalHbO * MICROMOLAR_CONV, 'g', 'LineWidth', 1.5);
hold on;

% Marking the peak with a red star and an exact text label (e.g. 57.6 BPM)
plot(heartBeatFreq, heartBeatStrength * MICROMOLAR_CONV, 'r*', 'MarkerSize', 8,
'LineWidth', 1.5);
text(heartBeatFreq + 0.1, heartBeatStrength * MICROMOLAR_CONV, sprintf('Peak: %.2f Hz
(%.1f BPM)', heartBeatFreq, bpm), 'Color', 'red', 'FontSize', 10, 'FontWeight',
'bold');

xlim([0.2, max(freqAxis)]); % Starting from 0.2Hz to exclude slow artifacts
title('Concentration as function of frequency for channel 1');
xlabel('Frequency (Hz)')
ylabel('\Delta Concentration (\muM)');
legend('\Delta HbO', 'Heartbeat Peak', 'location', 'best');
grid on;
hold off;

% Automatically exporting Question 3 Frequency Domain plot at 600 DPI
exportgraphics(fig_fft, 'FFT_Spectrum_Channel1.png', 'Resolution', 600);
```

```
% Printouts
fprintf('Heartbeat Signal Strength: %.4e\n', heartBeatStrength);
fprintf('Baseline Noise Level:      %.4e\n', noiseAvg);
fprintf('Heart rate (BPM):          %.4f\n', bpm);
fprintf('Final SNR:                 %.4f\n', SNR);
```

File: CalcNIRS.m

Matlab

```
% Dan Sadeh 314631763
function [dHbR, dHbO, fig] = CalcNIRS(dataFile, sourceDetectorSep, tissueType,
plotChannelIdx, ...
    extinctionCoefficientsFile, DPFperTissueFile, relDPFfile)

    % ------------------------------------------------------------------------
    % Description : The function calculates and plots HbR and HbO concentration
    %               changes using the Modified Beer-Lambert Law.
    % Assumptions : The data matrix 'd' contains exactly 40 columns (20 channels
    %               per wavelength). sourceDetectorSep is positive.
    % Inputs :
    %   dataFile - .mat file with intensity data (d), time (t), and Lambda.
    %   sourceDetectorSep - Source-Detector Separation distance in cm.
    %   tissueType - String matching a row in the DPF text file.
    %   plotChannelIdx - Vector indicating channels to plot.
    %   extinctionCoefficientsFile - .csv file with extinction parameters.
    %   DPFperTissueFile - .txt file with baseline DPF multipliers.
    %   relDPFfile - .csv file for wavelength-specific DPF adjustment.
    % Outputs :
    %   dHbR - HbR concentration change matrix (time x 20).
    %   dHbO - HbO concentration change matrix (time x 20).
    %   fig  - Handle to generated figures. Empty if no plots requested.
    % ------------------------------------------------------------------------

    %% Calculation segment - Input Validation and Constants
    fig = [];

    % Defining hardware architecture constants to eliminate magic numbers
    NUM_CHANNELS = 20;
    NUM_COLUMNS = 40;
    MICROMOLAR_CONV = 1e6;

    if nargin < 3
        error('You must provide at least dataFile, sourceDetectorSep, and
tissueType.');
    end

    if ~(ischar(dataFile) || isstring(dataFile)) || ~exist(dataFile, 'file')
        error('dataFile must be a valid text string pointing to an existing file.');
    end
```

```
    if ~isnumeric(sourceDetectorSep) || ~isscalar(sourceDetectorSep) ||
sourceDetectorSep <= 0
        error('sourceDetectorSep must be a single positive number.');
    end

    if nargin < 4 || isempty(plotChannelIdx)
        plotChannelIdx = [];
    elseif ~isnumeric(plotChannelIdx) || ~isequal(round(plotChannelIdx),
plotChannelIdx) || any(plotChannelIdx < 1) || any(plotChannelIdx > NUM_CHANNELS)
        error('plotChannelIdx must contain only whole numbers between 1 and %d.',
NUM_CHANNELS);
    end

    if nargin < 5 || isempty(extinctionCoefficientsFile); extinctionCoefficientsFile
= 'ExtinctionCoefficientsData.csv'; end
    if nargin < 6 || isempty(DPFperTissueFile); DPFperTissueFile =
'DPFperTissue.txt'; end
    if nargin < 7 || isempty(relDPFfile); relDPFfile = 'RelativeDPFCoefficients.csv';
end

    %% Calculation segment - Data Extraction
    dataStruct = load(dataFile);

    if ~isfield(dataStruct, 'd') || ~isfield(dataStruct, 't') || ~isfield(dataStruct,
'SD') || ~isfield(dataStruct.SD, 'Lambda')
        error('Loaded file is missing required fields (d, t, SD, or Lambda).');
    end

    if size(dataStruct.d, 2) ~= NUM_COLUMNS
        error('Expected the data matrix to have exactly %d columns.', NUM_COLUMNS);
    end

    intensityMat = dataStruct.d;
    timeVec = dataStruct.t;
    lambda1 = dataStruct.SD.Lambda(1);
    lambda2 = dataStruct.SD.Lambda(2);

    extCoeffTable = readtable(extinctionCoefficientsFile);
    dpfTissueTable = readtable(DPFperTissueFile);
    relDpfTable = readtable(relDPFfile);

    %% Calculation segment - MBLL Pre-calculations
    % Extracting the tissue-specific pathlength multiplier
    tissueIdx = find(strcmp(dpfTissueTable.Tissue, tissueType));
    if isempty(tissueIdx)
        error('Tissue type not found in DPF table.');
    end
    baseDPF = dpfTissueTable.DPF(tissueIdx);

    % Adjusting pathlength based on photon scattering variance at different
wavelengths
    lambda1Idx = find(relDpfTable.wavelength == lambda1);
    lambda2Idx = find(relDpfTable.wavelength == lambda2);
    relDPF1 = relDpfTable.relDPFcoeff(lambda1Idx);
```

```
    relDPF2 = relDpfTable.relDPFcoeff(lambda2Idx);

    % Calculating Effective Pathlength (Leff)
    leffL1 = sourceDetectorSep * baseDPF * relDPF1;
    leffL2 = sourceDetectorSep * baseDPF * relDPF2;

    % Defining the inverse (1/Leff) here optimizes loop performance later
    invLeffL1 = 1 / leffL1;
    invLeffL2 = 1 / leffL2;

    % Building and immediately inverting the Extinction Coefficient Matrix (Epsilon)
    epsIdxL1 = find(extCoeffTable.wavelength == lambda1);
    epsIdxL2 = find(extCoeffTable.wavelength == lambda2);

    epsMatrix = [extCoeffTable.HbO2(epsIdxL1), extCoeffTable.HHb(epsIdxL1);
                 extCoeffTable.HbO2(epsIdxL2), extCoeffTable.HHb(epsIdxL2)];
    invEpsMatrix = inv(epsMatrix);

    %% Calculation segment - Core MBLL Matrix Multiplication
    % Preallocating arrays limits dynamic memory expansion inside the loop,
    % significantly reducing computational overhead.
    dHbO = zeros(length(timeVec), NUM_CHANNELS);
    dHbR = zeros(length(timeVec), NUM_CHANNELS);

    for channelIdx = 1:NUM_CHANNELS
        % Isolating baseline and continuous data for the specific channel
        intensityZeroL1 = intensityMat(1, channelIdx);
        intensityTimeL1 = intensityMat(:, channelIdx);

        intensityZeroL2 = intensityMat(1, NUM_CHANNELS + channelIdx);
        intensityTimeL2 = intensityMat(:, NUM_CHANNELS + channelIdx);

        % Calculating Optical Density using base 10 logarithmic attenuation
        opticalDensityL1 = log10(intensityZeroL1 ./ intensityTimeL1);
        opticalDensityL2 = log10(intensityZeroL2 ./ intensityTimeL2);

        % Normalizing optical density by the photon's travel distance
        scaledL1 = opticalDensityL1 .* invLeffL1;
        scaledL2 = opticalDensityL2 .* invLeffL2;

        % Solving the linear system to extract distinct HbO/HbR chromophore
concentrations
        rightSideMatrix = [scaledL1, scaledL2]';
        concentrations = invEpsMatrix * rightSideMatrix;

        dHbO(:, channelIdx) = concentrations(1, :)';
        dHbR(:, channelIdx) = concentrations(2, :)';
    end

    %% Plot segment - Data Visualization
    if isempty(plotChannelIdx)
        disp('No channels were given so no plot is generated.');
    else
        for ii = 1:length(plotChannelIdx)
            channelToPlot = plotChannelIdx(ii);
```

```
            fig(ii) = figure;

            % Multiplying by the conversion constant brings the values into a
readable microMolar integer range
            plot(timeVec, dHbO(:, channelToPlot) * MICROMOLAR_CONV, 'r', 'LineWidth',
1.5);
            hold on;
            plot(timeVec, dHbR(:, channelToPlot) * MICROMOLAR_CONV, 'b', 'LineWidth',
1.5);

            xlim([0, max(timeVec)]);
            title(sprintf('Concentration changes as func of time for channel %d',
channelToPlot));
            xlabel('Time (seconds)');
            ylabel('\Delta Concentration (\muM)');
            legend('\Delta HbO', '\Delta HbR', 'Location', 'best');
            grid on;
            hold off;
        end
    end
end
```